

Ten Stems, Twelve Branches: Why Only Sixty Pairings Remain

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2026-10-04 · 预印本 (preprint) · 未经同行评议 · v3

DOI: 10.5281/zenodo.22990186

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Abstract

Ten heavenly stems paired with twelve earthly branches, two at a time, ought to give one hundred and twenty arrangements. The almanac uses sixty. Every one of the sixty that is missing fails the same single rule: a yang stem pairs only with a yang branch, and a yin stem only with a yin branch. The same count of sixty follows independently from the least common multiple of ten and twelve, and from the gearing image used in teaching, in which the stems complete six turns and the branches five before the two return to 甲子 together. The round is called 一甲子, and also 六十甲子, the sixty-term cycle, and 花甲; one proposed derivation of that last name refers to the interleaving of odd with odd and even with even. 花甲之年 denotes sixty 虚岁, that is, nominal age, and falls close to fifty-nine completed years. The sixty are unwieldy as a single list and are cut into six sections of ten, each named after its own first member; two branches are left over in every section, and those are the 旬空, the void in the decade. The sixty need not be memorised. What must be memorised is the six section heads. The note is descriptive; it contains no predictive claims.

Keywords: heavenly stems; earthly branches; yin and yang; the sixty-term cycle; least common multiple; ten-day period; huajia; the void in the decade

1. There is only one rule

First separate the two series by yin and yang.

甲, 丙, 戊, 庚 and 壬 are the yang stems; 乙, 丁, 己, 辛 and 癸 are the yin stems. Among the branches, 子, 寅, 辰, 午, 申 and 戌 are yang; 丑, 卯, 巳, 未, 酉 and 亥 are yin.

In pairing, a yang stem takes only a yang branch, and a yin stem only a yin branch.

甲子 holds: 甲 is a yang stem and 子 a yang branch, the two in the same column. Change it to 甲丑 and it no longer holds. 甲 is still yang, but 丑 is yin, and the two cannot be brought together.

The rule is put as 奇配奇、偶配偶, "odd pairs with odd, even with even", and a figure follows it: 如齿轮咬合, 不可错位, "like gears in mesh, no displacement permitted".

Of the ten stems, every one can be paired with only six branches. Five yang stems against six yang branches, and five yin stems against six yin branches; multiply the two sides and the answer is sixty.

六十甲子

阳配阳，阴配阴

	子	丑	寅	卯	辰	巳	午	未	申	酉	戌	亥
甲	■		■		■		■		■		■	
乙		■		■		■		■		■		■
丙	■		■		■		■		■		■	
丁		■		■		■		■		■		■
戊	■		■		■		■		■		■	
己		■		■		■		■		■		■
庚	■		■		■		■		■		■	
辛		■		■		■		■		■		■
壬	■		■		■		■		■		■	
癸		■		■		■		■		■		■

朱砂格=成立的组合，共六十个
空白的六十格，被阴阳这条规矩剔除了

2. Another way of counting, and still sixty

The first count is the plainest. Ten times twelve is one hundred and twenty; the yin-yang rule cuts away half; sixty remain.

The second count needs one number: the least common multiple of ten and twelve, which is exactly sixty.

Take the stems and the branches as two gears with different numbers of teeth, set turning at the same moment.

The stems turn once, from 甲 to 癸, ten signs. The branches turn with them, from 子 to 酉, also ten. At this point the stems are used up, while two branches, 戌 and 亥, have not been paired off.

So the stems begin again from the top: 甲 takes 戌, 乙 takes 亥. Further on, 甲申, 乙酉, and so down the line.

The stems turn six times, sixty; the branches turn five times, also sixty. The two return to the starting point 甲子 at the same moment.

The sixty combinations complete one round, and then begin again, round upon round.

3. What this round is called

The round is called 一甲子, and also 六十甲子. Among ordinary people there is another name: 花甲.

Where 花甲 comes from has one explanation.

That explanation derives the name from 花着排, "interleaved". In his essay 说「花甲」 ("On huajia", Handan Daily, 28 January 2018), Li Zefu of the Handan Municipal Society for the Study of Chinese Culture gives it thus: in pairing stems with branches, an odd number takes an odd and an even takes an even, as in 甲子 and 乙丑, and never 甲丑 or 乙子; this is called 花着排, and from it the name 花甲 arises.

Apart from the name, one thing deserves separate mention: 花甲之年 means sixty 虚岁, and stands nearly a year away from the sixty years usually meant.

In the same essay Li Zefu singles this out: people who call themselves 花甲之年 and set out to celebrate a 花甲 birthday are in fact marking sixty completed years. He calls this "a misunderstanding caused by not understanding the Chinese tradition of counting years".

Nominal age is counted from birth as one, and gains one at every lunar new year. Sixty 虚岁 therefore falls at fifty-nine completed years.

4. Sixty combinations cut into six

Sixty combinations are too long a list, so they are cut. The cut is one section per ten.

The first section runs from 甲子 to 癸酉, exactly ten days, and is called a 旬. A section takes its name from its own first member, so this one is the 甲子 旬.

Every ten thereafter forms a 旬 in turn: 甲戌旬, 甲申旬, 甲午旬, 甲辰旬, 甲寅旬. The six sections together make exactly sixty.

旬	Positions	Range
甲子旬	1-10	甲子 to 癸酉
甲戌旬	11-20	甲戌 to 癸未
甲申旬	21-30	甲申 to 癸巳
甲午旬	31-40	甲午 to 癸卯
甲辰旬	41-50	甲辰 to 癸丑
甲寅旬	51-60	甲寅 to 癸亥

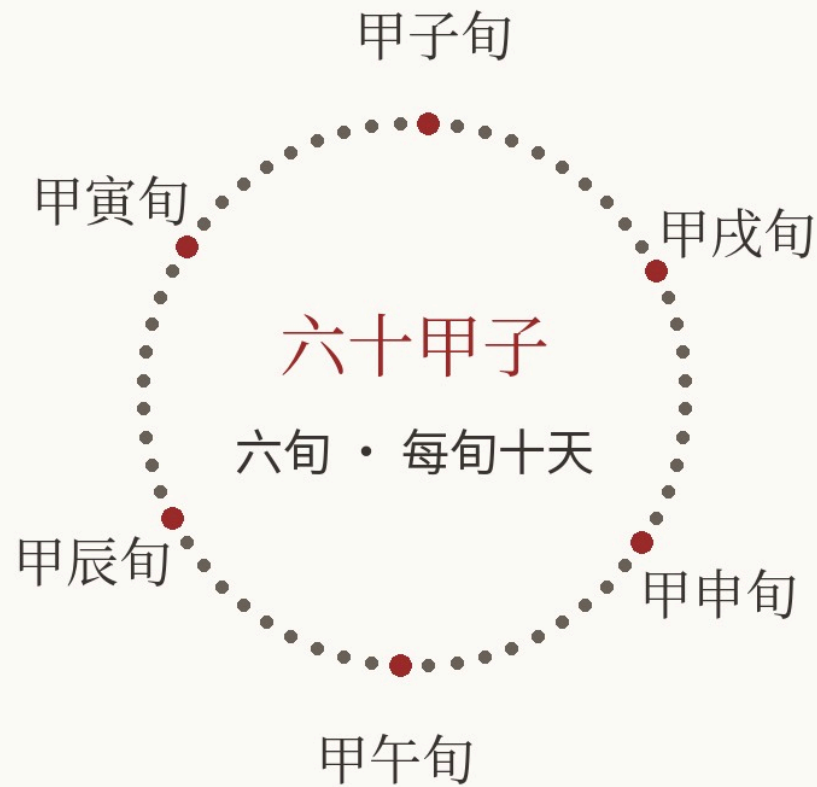
One thing is hidden in that table. Every 旬 pairs ten stems against ten branches, but there are twelve branches, so two of them must go unused.

In the 甲子 旬 the two passed over are 戌 and 亥; in the 甲戌 旬 they are 申 and 酉; and so on to the 甲寅 旬, where 子 and 丑 are passed over. The two branches thus wanting are called 旬空.

How the 旬空 are taken, and what use is made of them, is another course of study. Here we stop at the name.

One word on the word: this 旬 is the same 旬 as in 一句十天, the 旬 of 上旬, 中旬 and 下旬, the first, middle and last third of a month.

六旬
六十组，每十组一旬



每旬十个天干配十个地支，地支有十二个
必有两个轮不上

Closing

The sixty combinations laid out, and the rule is one: yang with yang, yin with yin. The two gears turn six times and five, and come back to where they started. The sixty are cut into six sections of ten days, named each from its head.

The sixty combinations need not be learned by heart. What must be learned by heart is the six section heads.

A&Q

Why is there no 甲丑?

甲 is a yang stem and 丑 a yin branch. The rule of pairing is that a yang stem takes only a yang branch and a yin stem only a yin branch; no crossing is allowed. 甲丑 breaks this rule, so it does not hold.

How are the sixty counted out?

Ten times twelve gives one hundred and twenty arrangements; the yin-yang rule removes half, and sixty remain. Or take the least common multiple: the least common multiple of ten and twelve is sixty, and the stems turn six times while the branches turn five, returning to the starting point together.

Where does the name 花甲 come from?

From 花着排, that is, in pairing stems and branches an odd takes an odd and an even takes an even, as in 甲子 and 乙丑, and never 甲丑 or 乙子. The explanation is given by Li Zefu in 说「花甲」 (Handan Daily, 28 January 2018).

Is 花甲之年 sixty completed years?

It is sixty 虚岁. Nominal age counts from birth as one and adds one at every lunar new year, so it stands nearly a year away from completed years.

What is 旬空?

The sixty combinations are one 旬 per ten, six 旬 in all. Each 旬 pairs ten stems with ten branches, and with two branches to spare, two of them must go unused. The two wanting are called 旬空. How they are taken and how they are used is another course of study.

Note by Chujia Kuang

The sixty combinations need not all be committed to memory. Learn the six section heads first: 甲子, 甲戌, 甲申, 甲午, 甲辰, 甲寅. One section a day, and in six days they are done. Once these six are held, every later exercise that comes round to a 旬 has somewhere to stand.

About the author

Chujia Kuang (邝楚嘉), pen name Jiayan Yide (嘉言一得), is an independent researcher in Jieyang, Guangdong, devoted to archiving verifiable primary sources and calendar data.

This column writes about calendrics, solar terms, the stems and branches and orientation, and also about what he reads. The pen name 嘉言一得 comes from the 史记: 愚者千虑, 必有一得, "the foolish man's thousand thoughts must yield one gain". Where something is written wrongly, corrections are welcome.

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Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Data availability. No new data were created or analysed in this study.

Language. The original Chinese text is deposited as a companion file to this record (十个字配十二个字，为什么只剩六十组).

References

[1] Li Zefu. Shuo "Huajia" [说「花甲」][N]. Handan Daily, 2018.